

ALAR, China

WMO: 517300

Lat: 40.550N

Lon: 81.267E

Elev: 3323

StdP: 13.01

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	4.0	-4.8	4.8	9.3	-1.6	5.7	11.1	9.6	27.8	8.3	25.8	1.2	220	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	94.5	67.8	92.4	66.7	90.2	65.9	72.6	88.7	70.6	86.9	68.9	85.4	3.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	113.9	83.7	65.1	105.1	80.7	63.2	98.3	78.1	38.8	88.9	36.9	87.1	35.3	85.6	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.5	10.8	8.9	-1.7	98.9	4.4	2.3	-4.9	100.5	-7.5	101.9	-9.9	103.1	-13.2	104.8		
			-2.2	77.0	4.4	3.2	-5.3	79.3	-7.9	81.1	-10.3	82.9	-13.5	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	18.7	30.0	46.3	60.3	69.1	74.8	76.6	74.7	66.5	53.2	37.9	23.7
	DBStd	20.87	5.50	6.76	7.22	6.66	5.78	4.48	4.14	4.32	4.68	6.14	6.10	5.05
	HDD50	2918	970	561	160	8	0	0	0	0	0	37	365	817
	HDD65	5689	1435	981	578	168	28	1	0	0	37	365	814	1282
	CDD50	3923	0	0	47	316	593	743	824	765	496	138	1	0
	CDD65	1219	0	0	0	26	156	294	359	300	83	1	0	0
	CDH74	14473	0	0	10	550	1982	3393	4102	3292	1060	84	0	0
	CDH80	6018	0	0	0	143	733	1504	1933	1413	285	7	0	0
Wind	WSAvg	3.3	2.2	2.7	3.5	4.5	4.7	4.3	3.9	3.5	3.1	2.6	2.1	2.1
Precipitation	PrecAvg	2.4	0.1	0.0	0.0	0.1	0.3	0.5	0.5	0.5	0.3	0.1	0.0	0.0
	PrecMax	4.2	0.4	0.4	0.4	0.3	1.3	1.3	1.1	2.2	1.3	0.9	0.3	0.6
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
	PrecStd	0.8	0.1	0.1	0.1	0.1	0.3	0.3	0.3	0.5	0.4	0.2	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	53.8	75.0	88.6	93.4	96.0	98.3	96.1	89.3	80.5	65.1	44.4
		MCWB	30.3	39.0	49.9	57.3	60.2	66.0	69.2	70.5	65.3	56.8	45.9	34.8
	2%	DB	36.4	50.0	70.2	84.3	90.3	93.4	95.1	92.8	86.1	76.2	60.2	39.9
		MCWB	27.9	37.0	47.7	55.5	59.4	65.1	69.6	69.3	63.6	54.8	43.6	31.3
	5%	DB	33.1	46.9	66.1	80.7	87.6	91.2	92.5	90.4	83.4	72.3	56.3	36.4
		MCWB	25.9	35.2	46.1	53.8	58.8	65.0	68.7	68.6	62.7	53.0	42.0	29.3
	10%	DB	30.1	43.6	62.0	76.6	84.6	88.7	89.7	87.8	80.4	68.5	51.0	33.2
		MCWB	24.0	33.1	44.5	52.0	57.8	64.0	67.9	67.9	61.8	51.4	39.7	27.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.5	40.5	50.7	58.4	64.4	72.2	75.9	75.6	70.9	59.6	46.9	35.8
		MCDB	39.9	50.9	72.5	86.2	84.6	89.7	92.1	89.7	83.8	71.9	63.0	43.5
	2%	WB	28.1	37.7	48.4	56.1	61.5	68.8	72.7	72.9	67.2	57.2	44.1	31.8
		MCDB	35.9	48.8	68.0	82.2	85.1	87.9	89.6	88.1	80.9	72.2	58.8	39.0
	5%	WB	26.2	35.7	46.7	54.4	60.0	66.7	70.7	70.7	64.9	55.1	42.2	29.5
		MCDB	32.7	46.2	64.8	78.6	83.4	86.1	87.8	86.1	78.7	69.6	55.3	35.8
	10%	WB	24.2	33.4	44.8	52.7	58.6	65.2	69.1	69.0	63.0	52.7	40.1	27.5
		MCDB	29.8	43.0	61.1	74.6	81.0	84.2	86.0	84.6	76.8	65.7	50.6	32.9
Mean Daily Temperature Range	5% DB	MDBR	23.9	24.8	24.7	25.6	24.9	24.1	23.3	23.7	26.8	31.0	26.4	21.0
		MCDBR	27.5	28.0	30.2	33.8	31.3	29.1	29.1	29.0	31.9	34.7	32.8	24.6
		MCWBR	19.8	16.4	12.9	11.8	9.6	10.2	10.9	12.7	14.3	15.3	17.3	16.7
	5% WB	MCDBR	26.7	26.8	28.9	30.8	27.8	26.0	25.5	26.5	28.0	31.6	32.2	23.7
		MCWBR	19.3	15.9	12.6	11.2	9.5	10.8	12.1	13.8	14.4	14.7	17.1	16.2
Clear-Sky Solar Irradiance	taub	0.361	0.426	0.538	0.601	0.628	0.660	0.610	0.657	0.660	0.547	0.447	0.385	
	taud	2.033	1.894	1.654	1.558	1.561	1.546	1.661	1.569	1.527	1.672	1.855	1.964	
	Ebn at Noon	239	238	222	220	219	212	221	205	189	194	201	214	
	Edh at Noon	37	50	72	84	86	88	77	82	81	62	45	37	
All-Sky Solar Radiation	RadAvg	776	1073	1467	1829	2080	2161	2101	1874	1606	1285	902	698	
	RadStd	93	80	82	78	89	84	86	110	60	50	45	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.76	-2.00	-2.16	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	-1.37	-1.46	N/A	N/A	N/A	-92

Nomenclature: See separate page